

Asymmetric Capital Buffer Control

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1 Introduction

The question is how an asymmetric penalty on a capital shortfall shapes a bank's recapitalisation policy and the volatility of its capital buffer. The model treats recapitalisation as impulse control under a fixed issuance cost and a regulatory cap on risky exposure.

Table 1: Headlines.

ID	Headline	Model rows	Scope note
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2 Model

Table 2: Model results.

ID	Claim	Statement	Lean	Proof
M1	The deficit-to-surplus variance ratio of the discrete-time recursion is λ_V^4 , for every saturation scale.	$V_0 > 0, \theta > 0, \lambda_V > 0,$ $V(z) = V_0 \exp(-2 \log \lambda_V \tanh(z/\theta)) \Rightarrow$ $V(-z)/V(z) \rightarrow \lambda_V^4$ as $z \rightarrow \infty$.	<code>RateBased.Vresp_ratio_</code> <code>eq, RateBased.tendsto_</code> <code>tanh_atTop, RateBased.</code> <code>saturated_ratio_</code> <code>tendsto, RateBased.</code> <code>saturated_ratio_</code> <code>lambda4</code>	p. 4
M2	The solvency branch of the cap binds exactly up to the crossing point $\bar{x}$, and there the z -volatility is the constant v_1^2 .	$0 < a_1 < a_3, a_2 < 1 \Rightarrow$ $\bar{x} - 1 = a_1(1 - a_2)/(a_3 - a_1) > 0,$ $u(x) = (x - 1)/a_1$ for $x \leq \bar{x}, u(x) = (x - a_2)/a_3$ for $x \geq \bar{x}$. If also $q > 0, 1 < x \leq \bar{x}$ and $\pi(x)x = u(x)$, then $\zeta(x) = v_1^2$.	<code>CapGeometry.</code> <code>hasDerivAt_zmap,</code> <code>CapGeometry.zvol2_eq,</code> <code>CapGeometry.xbar_sub_</code> <code>one, CapGeometry.one_</code> <code>lt_xbar, CapGeometry.</code> <code>solvency_binds,</code> <code>CapGeometry.liquidity_</code> <code>binds, CapGeometry.</code> <code>zvol2_solvency</code>	p. 4
M3	For any exposure policy that binds the cap near an end of the state space, the z -volatility tends to v_1^2 at the distress boundary and to v_3^2 deep in surplus.	$q > 0, 0 < a_1 < a_3, a_2 < 1$. If $\pi(x)x = u(x)$ for all x near 1^+ then $\zeta(x) \rightarrow v_1^2$ as $x \rightarrow 1^+$. If $\pi(x)x = u(x)$ for all large x then $\zeta(x) \rightarrow v_3^2$ as $x \rightarrow \infty$. Both fail without $a_1 < a_3$ or without binding.	<code>CapGeometry.deficit_</code> <code>limit, CapGeometry.</code> <code>surplus_limit,</code> <code>CapGeometry.variance_</code> <code>ratio_is_cap_geometry,</code> <code>CapGeometry.control_</code> <code>surplus_limit_needs_</code> <code>a1_lt_a3, CapGeometry.</code> <code>control_constancy_</code> <code>needs_binding</code>	p. 4

3 Literature

Table 3: Papers read.

ID	Paper	What we rely on	Quote	Page	Rows
L1	Bayraktar et al. [2026]	The cap u , the diffusion $\sigma^2(x, \pi)$ and the crossing point $\bar{x}$ of M2 are this paper's model, its eq. (4) and its switching point, 1.168 at its baseline (p. 15). M2 and M3 are statements about its cap. The parallel ends there, since the paper evaluates no limit of the volatility and has no asymmetry parameter.	"the parameter a_3 scales the entire liquidity constraint and determines its asymptotic upper bound"	17	M2, M3, C1, C2
L2	Barberis et al. [2001]	With a constant level of loss aversion, the volatility of log returns equals that of dividend growth for every level (12 percent at four values of b_0 , their Table XIII), the parallel to the silence of λ_S in C2. The parallel ends at the mechanism, a constant price-dividend ratio in an endowment economy rather than a binding regulatory cap.	"the volatility of log returns in this model is equal to the volatility of log dividend growth, namely 12 percent."	47	C2
L3	Engle and Siriwardane [2018]	They split observed equity volatility asymmetry into a mechanical-leverage share (3% by their correlation measure) and a market-exposure share, separating a structural source from a non-structural one as C2 does. The parallel ends at the object, since their asymmetry is a response of variance to the sign of news and not a ratio of variances across regions of a state variable.	"Volatility asymmetry is mostly explained by exposure to the aggregate market, not a mechanical leverage effect."	449	C2
L4	Li et al. [2023]	In an unconstrained loss-averse consumption and investment problem, the large-wealth limits of the optimal controls depend on the preference parameters, the contrast to the silence of λ_S in C2. The parallel ends at the constraint, since their risky position is unconstrained and the limit is a Merton-type scaling, while in C2 a binding regulatory cap fixes it. By their own formulas the loss-aversion degree enters those limits only when the curvatures on gains and losses are equal.	"our asymptotic limits differ significantly from the ones in the Merton's problem, which now sensitively depends on the reference degree parameter"	13	C2

Every paper the tables name has been read.

4 Conclusions

Table 4 holds the conclusions that rest on model rows and on papers read in full. Table 5 holds every claim that a result is new and every other claim still waiting, and each of its rows names what it waits on.

Table 4: Conclusions.

ID	Statement	Rows
C1	Wherever the optimal policy binds the regulatory cap near both ends of the state space, the saturation limits of the z -volatility are v_1^2 and v_3^2 . Their ratio involves $(\sigma, \sigma_L, a_1, a_3, c)$ only, so read as the variance ratio λ_V^4 it carries no information about λ_S .	M1, M2, M3, L1

Table 5: Pending conclusions.

ID	Statement	Rows
C2	That the saturation limits of the z -volatility in a regulated bank's capital problem carry no information about the asymmetry parameter is new. Among the papers read, Bayraktar et al. [2026] define the cap without evaluating these limits (L1), Barberis et al. [2001] find the level of loss aversion silent in return volatility through a constant price-dividend ratio rather than a binding cap (L2), and Engle and Siriwardane [2018] assign observed volatility asymmetry to market exposure rather than mechanical leverage, for a news response rather than a ratio of variances (L3), and Li et al. [2023] find preference parameters in the large-wealth limits of an unconstrained loss-averse control rather than a capped one (L4). <i>Pending</i> : a search for papers that evaluate limits of state volatility under a regulatory cap on the risky position.	M3, L1, L2, L3, L4

References

- Nicholas Barberis, Ming Huang, and Tano Santos. Prospect theory and asset prices. *The Quarterly Journal of Economics*, 116(1):1–53, 2001. doi: 10.1162/003355301556310.
- Erhan Bayraktar, Etienne Chevalier, Vathana Ly Vath, and Yuqiong Wang. Tractable bank capital structure: Optimal control under Basel III constraints, 2026. arXiv:2603.14557v2.
- Robert F. Engle and Emil N. Siriwardane. Structural GARCH: The volatility-leverage connection. *The Review of Financial Studies*, 31(2):449–492, 2018. doi: 10.1093/rfs/hhx099.
- Xun Li, Xiang Yu, and Qinyi Zhang. Optimal consumption with loss aversion and reference to past spending maximum, 2023. arXiv:2108.02648v4.

A Proofs

Notation. In discrete time $V(z)$ is the conditional variance of the shock at log buffer ratio z , with induced variance asymmetry $\lambda_V > 0$, saturation scale $\theta > 0$ and level $V_0 > 0$. In continuous time $x \in (1, \infty)$ is the ratio of assets to liabilities, $q > 0$ the ratio requirement, and $z(x) = \log((x-1)/q)$. The risky exposure is $\pi(x)$, and the regulatory cap gives the capped exposure $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with crossing point $\bar{x} = (a_3 - a_1 a_2)/(a_3 - a_1)$. The diffusion of x is $\sigma^2(x, \pi) = \pi^2 \sigma^2 x^2 + 2\pi c \sigma \sigma_L x(1-x) + \sigma_L^2 (1-x)^2$, with c the correlation between the risky return and the liability shock. The z -volatility is $\zeta(x) = z'(x)^2 \sigma^2(x, \pi(x))$, the exposure ratio is $\omega(x) = \pi(x)x/(x-1)$, and $v_i^2 = \sigma^2/a_i^2 - 2c\sigma\sigma_L/a_i + \sigma_L^2$ for $i \in \{1, 3\}$.

Proof of M1

Lean: RateBased.Vresp_ratio_eq, RateBased.tendsto_tanh_atTop, RateBased.saturated_ratio_tendsto, RateBased.saturated_ratio_lambda4.

For every z ,

$$\begin{aligned}\frac{V(-z)}{V(z)} &= \exp(4 \log \lambda_V \tanh(z/\theta)). \\ \tanh y &= 1 - \frac{2}{e^{2y} + 1} \rightarrow 1 \quad (y \rightarrow \infty), \quad z/\theta \rightarrow \infty \quad (z \rightarrow \infty, \theta > 0). \\ \exp(4 \log \lambda_V \tanh(z/\theta)) &\rightarrow \exp(4 \log \lambda_V) = \lambda_V^4.\end{aligned}$$

θ enters only through $z/\theta \rightarrow \infty$, so the limit is free of θ .

□

Proof of M2

Lean: CapGeometry.hasDerivAt_zmap, CapGeometry.zvol2_eq, CapGeometry.xbar_sub_one, CapGeometry.one_lt_xbar, CapGeometry.solvency_binds, CapGeometry.liquidity_binds, CapGeometry.zvol2_solvency.

$$\begin{aligned}\frac{x-1}{a_1} \leq \frac{x-a_2}{a_3} &\iff (a_3 - a_1)x \leq a_3 - a_1 a_2 \iff x \leq \bar{x}, \quad \bar{x} - 1 = \frac{a_1(1-a_2)}{a_3 - a_1} > 0. \\ z'(x) &= \frac{1}{x-1}, \quad \zeta(x) = \omega(x)^2 \sigma^2 - 2 \omega(x) c \sigma \sigma_L + \sigma_L^2. \\ 1 < x \leq \bar{x}, \pi(x)x = u(x) &\Rightarrow \omega(x) = \frac{1}{a_1} \Rightarrow \zeta(x) = v_1^2.\end{aligned}$$

□

Proof of M3

Lean: CapGeometry.deficit_limit, CapGeometry.surplus_limit, CapGeometry.variance_ratio_is_cap_geometry, CapGeometry.control_surplus_limit_needs_a1_lt_a3, CapGeometry.control_constancy_needs_binding.

Near 1^+ , $\bar{x} > 1$ by M2, so $1 < x < \bar{x}$ eventually and $\zeta(x) = v_1^2$ there. For $x \geq \bar{x}$,

$$\begin{aligned}\omega(x) &= \frac{x-a_2}{a_3(x-1)} = \frac{1}{a_3} + \frac{1-a_2}{a_3} (x-1)^{-1} \rightarrow \frac{1}{a_3}, \\ \zeta(x) &= \omega(x)^2 \sigma^2 - 2 \omega(x) c \sigma \sigma_L + \sigma_L^2 \rightarrow v_3^2.\end{aligned}$$

Controls, at $\sigma = 1$, $\sigma_L = c = 0$, $q = 1$.

$$\begin{aligned}a_1 = \frac{1}{2}, a_2 = 0, a_3 = \frac{1}{4} &\Rightarrow u(x) = 2(x-1), \zeta \equiv 4 \neq 16 = v_3^2. \\ \pi \equiv 0 &\Rightarrow \zeta(2) = 0 \neq 4 = v_1^2.\end{aligned}$$

□

B Evidence and verification

Toolchain. Lean v4.32.0-rc1 with Mathlib at tag v4.32.0-rc1, commit 360da6fa66c1, pinned in lean/mathlib/lake-manifest.json. Every file under lean/mathlib/ is a build root. A theorem passes the axiom audit when it depends on no axiom outside propext, Classical.choice and Quot.sound.

Checks. `verify.sh` runs `checks/verify_manuscript.py`, `checks/verify_mathlib.py` and `checks/verify_docs.py`, and writes their output to `VERIFICATION.md`. `lit/verify_lit.sh` runs the suite of every literature record.

What Lean does not carry. Two inputs enter the theorems as hypotheses or definitions. The z -volatility ζ is defined as $z'(x)^2 \sigma^2(x, \pi(x))$, and that this is the diffusion coefficient of $z(X_t)$ is Itô's formula, which is not proved (M2, M3). That the optimal policy binds the cap near both ends of the state space is a hypothesis of M3, and C1 states it as its condition. `PROOFS_v2.tex` supports it only by the numerical solve, which is evidence and not proof.

Illustrations. No row rests on SymPy or on numerical sampling.

C Terminology

Term	Symbol	Meaning
Induced variance asymmetry	λ_V	The asymmetry read off the variance ratio, distinct from λ_S .
Variance ratio	λ_V^4	The ratio of conditional variances deep in deficit to deep in surplus (M1).
Saturation scale	θ	How sharply the conditional variance moves between its deficit and surplus levels.
Asymmetry parameter	λ_S	The weight on shortfall below the reference in the objective.
Regulatory cap	$u(x)$	The capped exposure $\min\{(x-1)/a_1, (x-a_2)/a_3\}$.
Cap geometry	a_1, a_2, a_3	The parameters of the regulatory cap.
Solvency and liquidity regime	$x \leq \bar{x}, x \geq \bar{x}$	The ranges where the first and the second branch of u bind (M2).
z -volatility	$\zeta(x)$	The squared diffusion coefficient of z (Appendix B).
Saturation limits	v_1^2, v_3^2	The limits of ζ at the distress boundary and deep in surplus (M3).

D Primitives and scope

The regulatory cap. The capped exposure is $u(x) = \min\{(x-1)/a_1, (x-a_2)/a_3\}$ with $0 < a_1 < a_3$ and $a_2 < 1$, taken from `PROOFS_v2.tex`. Under these restrictions the solvency branch binds near the distress boundary and the liquidity branch deep in surplus (M2). With $a_1 \geq a_3$ the surplus limit of M3 fails, by its control. Rows M2, M3 and C1 depend on it.

The variance ratio in continuous time. `PROOFS_v2.tex` reads the discrete-time λ_V^4 of M1 in the continuous-time model as v_1^2/v_3^2 , identifying the conditional variance $V(z)$ deep in deficit and deep in surplus with the two saturation limits of ζ . This is a definition, not a result, and no theorem connects the discrete-time recursion to the continuous-time diffusion. C1 depends on it.

E Refuted conjectures